Sec: SR.IIT_*COSC(MODEL-A) | GTM-4 | Date: 21-12-25
Time: 3 Hrs | JEE-MAIN | Max. Marks: 300

KEY SHEET

PHYSICS

1	**B**	2	**A**	3	**C**	4	**D**	5	**D**
6	**D**	7	**C**	8	**A**	9	**A**	10	**D**
11	**B**	12	**C**	13	**B**	14	**C**	15	**A**
16	**D**	17	**C**	18	**D**	19	**C**	20	**A**
21	**60**	22	**12**	23	**6588**	24	**7**	25	**25**

CHEMISTRY

26	**B**	27	**D**	28	**C**	29	**A**	30	**C**
31	**B**	32	**A**	33	**B**	34	**C**	35	**B**
36	**C**	37	**D**	38	**A**	39	**C**	40	**C**
41	**C**	42	**B**	43	**C**	44	**D**	45	**C**
46	**12**	47	**5**	48	**24**	49	**1**	50	**9**

MATHEMATICS

51	**C**	52	**C**	53	**B**	54	**C**	55	**A**
56	**A**	57	**A**	58	**A**	59	**A**	60	**A**
61	**A**	62	**A**	63	**B**	64	**B**	65	**D**
66	**C**	67	**D**	68	**A**	69	**B**	70	**A**
71	**6084**	72	**2**	73	**15**	74	**14**	75	**22**

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SOLUTIONS
PHYSICS

1. $x^2 = Ka$

K is a positive constant

$$a = \frac{1}{K}x^2 \qquad \ldots\ldots(i)$$

$$\frac{da}{dx} = \frac{1}{K}\cdot 2x$$

$$\left(\frac{da}{dx}\right)_{(x=\sqrt{3})} = \frac{1}{K}\cdot 2\sqrt{3}$$

$$\tan 60^\circ = \frac{1}{K}\cdot 2\sqrt{3}$$

K = 2

From equation (i)

$$a(x=\sqrt{3}m) = \frac{1}{2}\cdot(\sqrt{3})^2 \qquad = 1.5\ m/s^2$$

2. $F_x = (4x - 8)\ N$

$$U(x) - U(x=0) = -\int_{x=0}^{x}(4x-8)\,dx$$

$$U(x) - 8 = \left[\frac{(4x-8)^2}{2(4)}\right]_0^x$$

$$U(x) = 8 - \frac{1}{8}\left[(4x-8)^2 - (-8)^2\right]$$

$$= 8 - \frac{16}{8}\left[(x-2)^2 - 4\right] = 8 - 2(x-2)^2 + 8 = 16 - 2(x-2)^2$$

U(x) – x curve is a parabola

U(x = 0) = 8 J

When U(x) = 0 $\Rightarrow x - 2 = \pm\sqrt{8}$, $x = x = 2 \pm 2\sqrt{2}$

U(x = 2 m) = 16 J

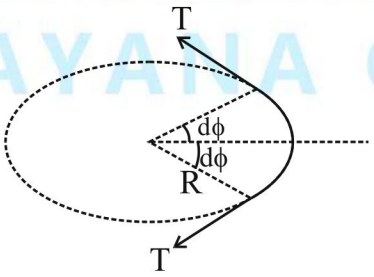
3. $2T\sin d\phi - dN\cos\theta = dm\omega^2 R$

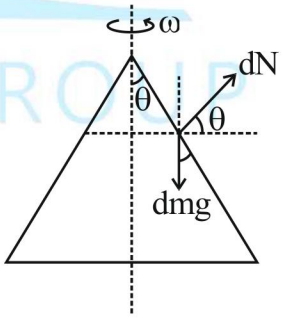
$dN\sin\theta = dmg$

$2T\sin d\phi = dmg\cot\theta + dm\omega^2 R$

$$2Td\phi \approx (g\cot\theta + \omega^2 R)\frac{m}{2\pi}(2d\phi)$$

$$T = \frac{mg}{2\pi}\left(\cot\theta + \frac{\omega^2 R}{g}\right)$$





4. $[P] = \frac{[MLT^{-2}]}{[L^2]} = [ML^{-1}T^{-2}]$

$[D] = [ML^{-3}]$

$[S] = \frac{[MLT^{-2}]}{[L]} = [MT^{-2}]$

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$[M^0L^0T^1] = [ML^{-1}T^{-2}]^a[ML^{-3}]^b\ [MT^{-2}]^c$

$a + b + c = 0$ ……(i)

$-a - 3b = 0 \quad \Rightarrow a = -3b$ ……(ii)

$-2a - 2c = 1 \quad \Rightarrow a + c = -\frac{1}{2}$ ……..(iii)

From (i) and (iii) $b = \frac{1}{2}$

From (ii) $a = -\frac{3}{2}$

$a = -\frac{3}{2},\ b = \frac{1}{2},\ c = 1$

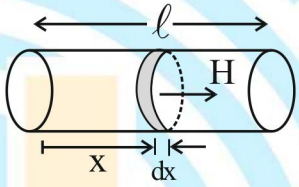
5. Equal number of nodes and antinodes occur in a closed pipe. 5 nodes and 5 antinodes are observed in 4th overtone

In a closed pipe 4th overtone is equivalent to $(2 \times 4 + 1)^{th}$ harmonic, i.e., 9th harmonic

6. $H = -kA\frac{dT}{dx}$

$Hdx = -\alpha TA\ dT$

$H\int_0^{\ell} dx = -\alpha A\int_{3T_0}^{T_0} T\,dT$ ….(i)



$H\int_0^{\ell/2} dx = -\alpha A\int_{3T_0}^{T} T\,dT$ ….(ii)

On solving this we get $T = \sqrt{5}\,T_0$

7. Let AB = ℓ,

RS = AS. tan θ

Q → c.m. of AB and CD

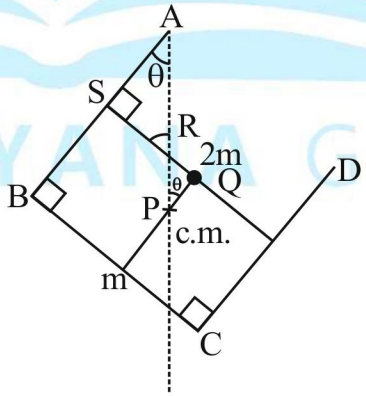
$RS = \frac{\ell}{2}\cdot\tan\theta$

$PQ = \left(\frac{m}{m+2m}\right)\frac{\ell}{2} = \frac{\ell}{6}$

$RQ = PQ.\tan\theta = \frac{\ell}{6}\tan\theta$

$SR + RQ = \frac{\ell}{2}$

$\frac{\ell}{2}\tan\theta + \frac{\ell}{6}\tan\theta = \frac{\ell}{2} \quad \Rightarrow \tan\theta = \frac{3}{4}$



8. $W_f = -\int_0^d \left\{\mu_0 + \mu_1\left(\frac{x}{2}\right)\right\} mg\,dx =$ $-mg\left[\mu_0 x + \frac{\mu_1}{d}\cdot\frac{x^2}{2}\right]_0^d =$

$-mgd\left(\mu_0 + \frac{\mu_1}{2}\right)$

Let v be the speed at the time of leaving the frictionless loop

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$$\frac{1}{2}mv^2 = \frac{1}{2}kx_0^2 - mgd\left(\mu_0 + \frac{\mu_1}{2}\right) - mgR$$

$$\frac{v^2}{2g} = \frac{kx_0^2}{2mg} - d\left(\mu_0 + \frac{\mu_1}{2}\right) - mgR$$

$$\frac{v^2}{2g} = \frac{kx_0^2}{2mg} - d\left(\mu_0 + \frac{\mu_1}{2}\right) - R$$

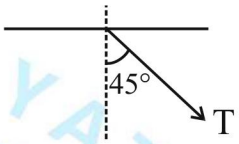
$$H_{max} = \frac{v^2}{2g} + R = \frac{kx_0^2}{2mg} - d\left(\mu_0 + \frac{\mu_1}{2}\right)$$

9. Let a be the acceleration of box, towards right just after release

System : Box

Frame: Earth

$T\sin 45° = ma \quad \Rightarrow T = \sqrt{2}\,ma$ .....(i)

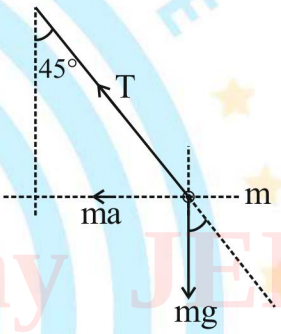


System : Ball

Frame: Box

$$T - mg\cos 45° + ma\cos 45° = \frac{m(0)^2}{L}$$

$$T - \frac{mg}{\sqrt{2}} + \frac{ma}{\sqrt{2}} = 0 \qquad \text{.....(ii)}$$



From equations (i) and (ii)

$$\sqrt{2}ma - \frac{mg}{\sqrt{2}} + \frac{ma}{\sqrt{2}} = 0$$

$$ma\left(\sqrt{2} + \frac{1}{\sqrt{2}}\right) = \frac{mg}{\sqrt{2}}$$

$$a = \frac{g}{3}$$

From equation (i)

$$T = \frac{\sqrt{2}mg}{3}$$

10. $$i(t) = 5\sin(100\pi t) + 5\sin\left(100\pi t + \frac{\pi}{6} + \frac{\pi}{2}\right) = 5\sin(100\pi t) + 5\sin\left(100\pi t + \frac{2\pi}{3}\right)$$

$$i_{max} = \sqrt{5^2 + 5^2 + 2(5)(5)\cos\frac{2\pi}{3}} = \sqrt{25 + 25 + 50\left(-\frac{1}{2}\right)} = 5\text{ A}$$

$$i_{rms} = \frac{i_{max}}{\sqrt{2}} = \frac{5}{\sqrt{2}}\text{A}$$

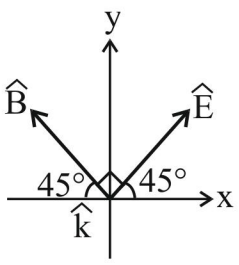
11. Unit vector along the direction of propagation is $\hat{k}$.

$\vec{E}$ is perpendicular to $\vec{B}$

$$\hat{k} = \hat{E} \times \hat{B}$$

$$\hat{k} = \left(\frac{\hat{x} + \hat{y}}{\sqrt{2}}\right) \times \hat{B}$$

$$\hat{B} = \frac{-\hat{x} + \hat{y}}{\sqrt{2}}$$



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12. $A_2 = 2A_1 \Rightarrow \frac{A_2}{A_1} = \frac{2}{1}$

$\frac{I_2}{I_1} = \frac{A_2^2}{A_1^2} = \frac{4}{1} \Rightarrow \frac{I_2}{4} = \frac{I_1}{4} = k$

$I_1 = k, \; I_2 = 4k$

$I_m = \left(\sqrt{k} + \sqrt{4k}\right)^2 = 9k$

$I = k + 4k + 2\sqrt{k}\sqrt{4k}\cos\phi$

$= 5k + 4k\cos\phi = \frac{9k}{9k}(5k + 4k\cos\phi)$

$= \frac{I_m}{9}\left[5 + 4\left(2\cos^2\frac{\phi}{2} - 1\right)\right] = \frac{I_m}{9}\left(1 + 8\cos^2\frac{\phi}{2}\right)$

13. $k_1 + k_2 = 5.5 \text{ MeV}$

$\vec{P}_1 + \vec{P}_2 = 0$

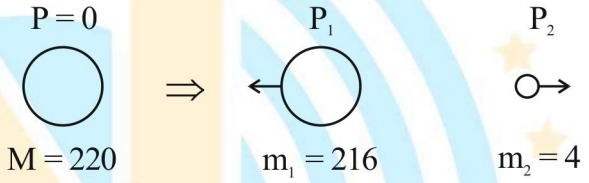
$-P_1 + P_2 = 0 \Rightarrow P_1 = P_2$

$\sqrt{2\times 216 k_1} = \sqrt{2\times 4k_2}$

$\frac{k_1}{2\times 4} = \frac{k_2}{2\times 216}$

$\Rightarrow \frac{k_1}{1} = \frac{k_2}{54} = \frac{k_1 + k_2}{1+54} = \frac{5.5\,\text{MeV}}{55}$

$k_2 = 54\times 0.1 \text{ MeV} = 5.4 \text{ MeV}$



14. $-\frac{1}{F_M} = 2\times\frac{1}{f_\ell} - \frac{1}{\infty}$

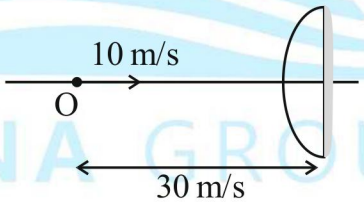
$F_m = -\frac{f_\ell}{2} = -\frac{10\,\text{cm}}{2}$

This combination will act like a concave mirror of focal length 5 cm.

$(V_{IM})_x = -m^2(V_{OM})_x$

$= -\left(\frac{F_M}{F_M - u}\right)^2 \times 10 \text{ cm/s}$

$= -\left(\frac{-5}{-5+30}\right)^2 \times 10 \text{ cm/s} = -\frac{1}{25}\times 10 \text{ cm/s}$



$(V_M)_x = 0 \Rightarrow (V_I)_x = -\frac{2}{5}\text{cm/s} = -0.4 \text{ cm/s}$

15. $E \propto r, \quad 0 \le r \le R$

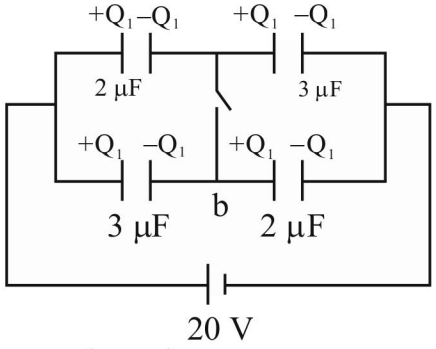
$\propto \frac{1}{r}, \quad r \ge R$

16. $1\times\lambda_1 = \frac{3\lambda_2 + 4\lambda_2}{2}$

$2\lambda_1 = 7\lambda_2$

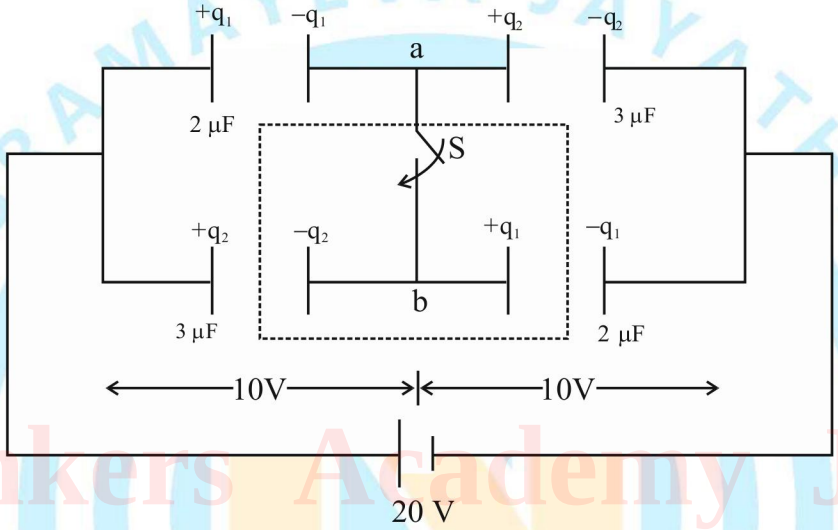
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17. Case I



$$Q_1 = \left(\frac{2\times 3}{2+3}\right)\times 20 = 24\ \mu C$$

Case II



$q_1 = 2\times 10 = +20\mu C$

$q_2 = 3\times 10\mu C = 30\mu C$

Charge passing through switch S from a to b

$= (q_1 - q_2) - (-Q_1 + Q_1) = -10\mu C$

Charge passing through switch S from b to a = 10 μC

18. By applying parallel combination of cells

$$E = \frac{\frac{2}{1+1}+\frac{4}{2}+\frac{4}{1+1}}{\frac{1}{2}+\frac{1}{2}+\frac{1}{2}} = \frac{10}{3}V = 3.33V$$

19. $i = \frac{5-0.7}{200}A = \frac{4.3}{200}A = 21.5mA$

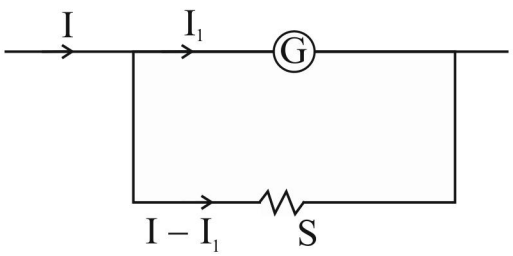
20. $(I - I_1)S = I_1G$

$$I_1 = \left(\frac{S}{S+G}\right)I$$

$$\frac{I_1}{I} = \frac{S}{S+G} \qquad .....(i)$$

$$\frac{1}{5} = \frac{4}{4+G} \Rightarrow 4+G = 20 \Rightarrow G = 16\Omega$$

Let $S' = 2\Omega$



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$$S_{eq} = \frac{2\times 4}{2+4}\Omega = \frac{4}{3}\Omega$$

$$\frac{I_1'}{I} = \frac{S_{eq}}{S_{eq}+G} = \frac{4/3}{\frac{4}{3}+16} = \frac{4}{52}$$

$$\frac{I_1'}{I} = \frac{1}{13}$$

Given $\frac{I_1}{I} = \frac{1}{5}$

$$\frac{I_1'}{I_1} = \frac{I_1'/I}{I_1/I} = \frac{5}{13}$$

21. $$B_P = 2\times\frac{\mu_0 I}{4\pi d}(\cos 0° + \cos 37°)$$

$$= \frac{2\times 4\pi\times 10^{-7}\times 1}{4\pi(1\times 10^{-2})\times\frac{3}{5}}\left[1+\frac{4}{5}\right] = 6\times 10^{-5}\ \text{T} = 60\ \mu\text{T}$$

22. Let I_0 be the intensity of incident light

Intensify of light after passing through

1st polaroid $= \frac{I_0}{2}$

$$I_2 = \frac{I_0}{2}\cos^2 30° = \frac{I_0}{2}\times\frac{3}{4} \quad ; \quad I_3 = \frac{I_0}{2}\times\frac{3}{4}\times\frac{3}{4}$$

Intensity of light after passing through 6th polaroid is

$$I_6 = \frac{I_0}{2}\left(\frac{3}{4}\right)^5$$

$$\frac{I_6}{I_0}\times 100 = \frac{243\times 100}{2\times 1024} = 11.86$$

Nearest integer = 12

23. $$\frac{1}{\lambda_1} = RZ^2\left(\frac{1}{1^2}-\frac{1}{\infty}\right) \quad ; \quad \frac{1}{\lambda_1} = RZ^2$$

$$\frac{1}{\lambda_2} = RZ^2\left(\frac{1}{2^2}-\frac{1}{3^2}\right) = \frac{1}{915}\left(\frac{9-4}{4\times 9}\right)$$

$$\lambda_2 = \frac{915\times 4\times 9}{5} = 183\times 36 = 6588\ \text{Å}$$

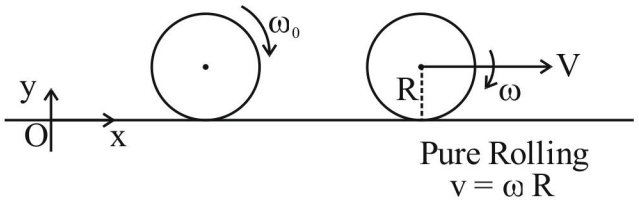
24. $\vec{\tau}_0^{ext} = 0$

$\vec{L}_0 = \text{constant}$

$$\frac{2}{5}MR^2\omega + RM(\omega R) = \frac{2}{5}MR^2\omega_0$$

$$\left(\frac{2}{5}+1\right)\omega = \frac{2}{5}\omega_0$$

$$\omega = \frac{2}{7}\omega_0 \quad ; \quad x = 7$$



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25. $\Delta U = Q - W = Q - \frac{Q}{5} = \frac{4Q}{5}$

$nC_V\Delta T = \frac{4Q}{5}$

$1 \times \frac{f}{2} R\ \Delta T = \frac{4Q}{5}$

$\frac{5}{2} R\ \Delta T = \frac{4Q}{5}$

$\frac{Q}{\Delta T} = \frac{25R}{8}$

$x = 25$

CHEMISTRY

26. $CO + \frac{1}{2}O_2 \rightarrow CO_2$

$CO_2 + 2KOH \rightarrow K_2CO_3 + H_2O$

Moles of KOH $= \frac{28}{56} = 0.50$

It corresponds to 0.25 mol of CO_2

Hence of CO = 1 – 0.25 = 0.75 ≡ mole of CO_2 formed

Mol. of KOH required = 2 × 0.75 = 1.5 = 1.5 × 56 = 84 g

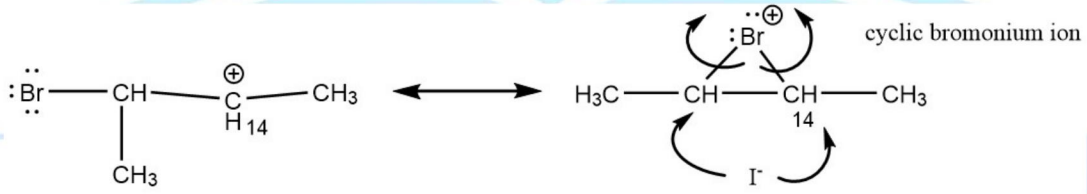
27. In S.I. units be P.E. $= \frac{-Ze^2}{4\pi\varepsilon_0 r}$. For Li^{2+}, Z = 3,

$\therefore$ P.E. $= \frac{-3e^2}{4\pi\varepsilon_0 r}$

28. In a period the value of ionisation potential increases from left to right with breaks where the atoms have some what stable configuration. In this case N has half filled stable orbital. Hence has highest ionisation energy. Thus the correct order is

B < C < O < N not B < C < N < O

29.



30. On determining hybridization from $H = \frac{1}{2}(V + M - C + A)$.

The hybridization of BF_3, $NO_2^{\ominus}$, $NH_2^{\ominus}$ and H_2O are sp^2, sp^2, sp^3 and sp^3 respectively.

31. The enthalpy of ionisation of weak acid is given by $\Delta H_{ion}(HA)$

$= \Delta H_{N(\text{weak acid/strong base})} - \Delta H_{N(\text{strong acid/strong base})}$

$= -56.1 - (-57.3) = 1.2$ kJ mol^{-1}

$\Delta H_{(\text{ionisation})} = 1.5$ kJ mol^{-1}

Hence % ionisation in 1M solution

$= \frac{(1.5 - 1.2)}{1.5} \times 100 = 20\%$

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32. $W_{bc} = +75$ J
$\Delta U_{ac} = \Delta U_{ab} + \Delta U_{bc}$
$-565 = 450 + \Delta U_{bc}$
$\Delta U_{bc} = -1015$ J
$Q_{bc} = \Delta U_{bc} - W_{bc}$
$= -1015 - 75 = -1090$ J

33. $2NOBr(g) \rightleftharpoons 2NO(g) + Br_2(g)$
$\frac{6P}{9}$ $\frac{2P}{9}$ $\frac{P}{9}$
Since total pressure is P
So, $\frac{6P}{9}+\frac{2P}{9}+\frac{P}{9}=P$

$$\therefore K_p = \frac{P_{NO}^2 \times p_{Br_2}}{p_{NOBr}^2} = \frac{\left(\frac{2P}{9}\right)^2 \times \frac{P}{9}}{\left(\frac{6P}{9}\right)^2}$$

$$\therefore \frac{K_p}{P} = \frac{1}{81}$$

34. Tellurium dioxide (TeO_2) acts as an oxidizing agent. This is because it can accept electrons and be reduced from its +4 oxidation state to a lower oxidation state.
Tellurium hydride (TeH_2) is considered acidic. This is due to its relatively low bond dissociation energy, which makes it prone to breaking, releasing protons (H^+)

35. (A) Group-13 size: B < Ga < Al < In ≃ TI
(B) In Group-13, Al has exceptional lower EN
EN : B > TI > In > Ga > Al
(C) $2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2$
$Al + HNO_3 \rightarrow$ No reaction (Due to protective oxide layer)
(D) +1 oxidation state is observed mainly for heavier Group-13 element like TI.
B and Al do not show +1 oxidation state
(E) $[Al(H_2O)_6]^{3+} \rightarrow$ Hybridisation : sp^3d^2

36. Factual

37. $[Ni(NH_3)_6]^{2+}$ has sp^3d^2 with unpaired e
$[Co(H_2O)_6]^{3+}$ has d^2sp^3 with 0 unpaired e
$[Cr(H_2O)_6]^{3+}$ has d^2sp^3 with 3 unpaired e
$[Mn(CN)_6]^{3-}$ has d^2sp^3 with 2 unpaired e

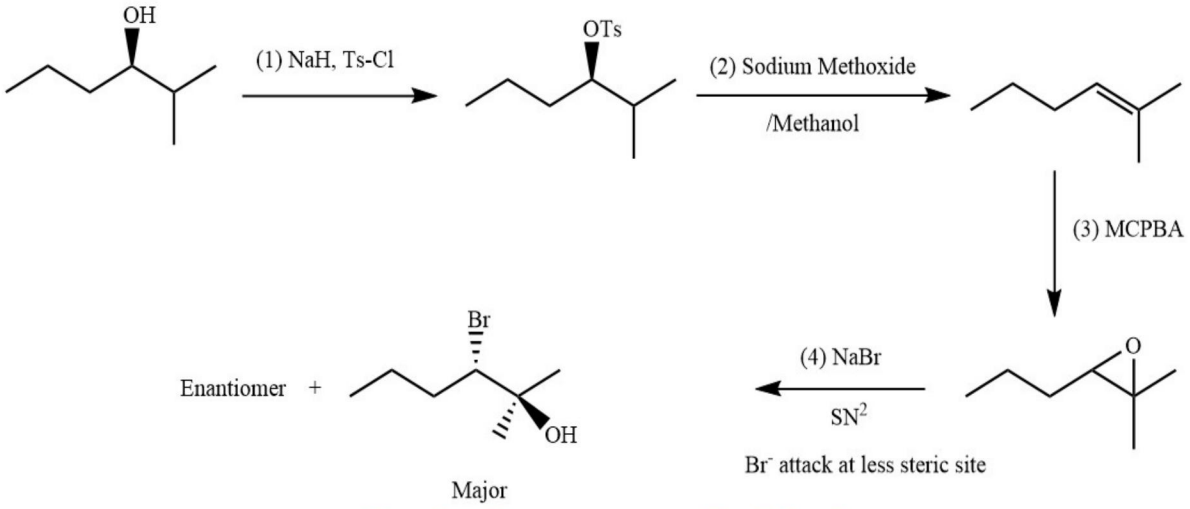
38. AgF is soluble in water while other AgX are insoluble
Agl is insoluble in aq. Ammonia solution
Br_2 is red gas
Cl– forms red fumes with $K_2Cr_2O_7 + H_2SO_4$

39. Th: $[Rn]5f^06d^27s^2$
Pu: $[Rn]5f^66d^07s^2$
Md: $[Rn]5f^{13}6d^07s^2$

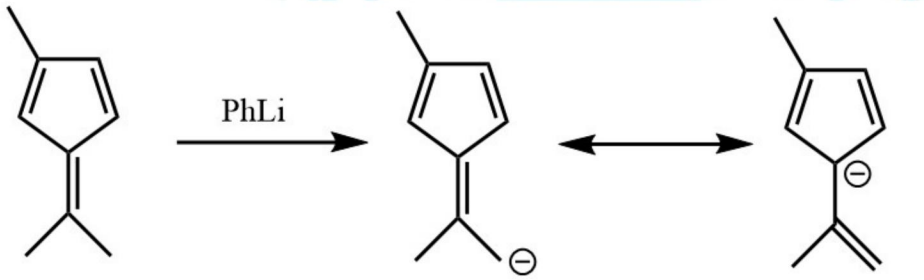
40. Sol. IUPAC name with stereo-descriptor

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41.

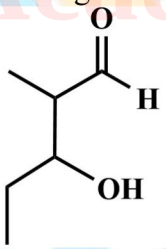


42.



43. Sol. Addition of HBr occurs via formation of stable benzylic carbocation.

44. Aldol with two different types of carbonyl can give self and cross products



this is self aldol product of the propanal which is not given as reactant

45. $[Co(ONO)(NH_3)_5Cl_2]$
Pentaamminenitrito-cobalt(III)chloride

46. For very dilute solutions

$$\frac{0.1\times10^{-3}\text{ mole } O_2}{1\text{L solution}} = \frac{0.1\times10^{-3}\text{ mol } O_2}{1\text{L water}} = \frac{0.1\times10^{-3}\text{ mol } O_2}{1000\text{g water}}$$

On applying Henry's Law

$$\frac{\frac{0.1\times10^{-3}}{1000}}{18} = \frac{p_{O_2}}{16\times10^5},\ p_{O_2} = \frac{160\times18}{1000}\text{ atm in air}$$

$$\text{Molar conc. of } O_2 \text{ in air} = \frac{P_{O_2}}{RT} = \frac{18\times16\times10^{-2}}{0.082\times300} = x$$

i.e. $100x = 11.69 \approx 12$

47. H_2, Li_2, C_2, N_2 and F_2 are diamagnetic
He_2^+, B_2, O_2 and O_2^- are paramagnetic

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48. $H_2O_2 \longrightarrow H_2O + \frac{1}{2}O_2$

Time	mole of H_2O_2	mole of O_2
0	a	0
20 min	a – x	0.5x
∞	0	0.5a

$U_{O_2} \propto$ mole of O_2

$0.5x \propto 16$ ml $\Rightarrow x \propto 32$ ml

$0.5a \propto V$ ml $\Rightarrow a \propto 2U$

For first order reaction

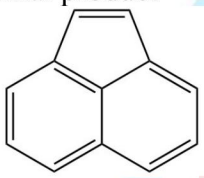
$$0.555 = \frac{2.303}{20}\log\frac{a}{a-x}$$

$$0.555 = \frac{2.303}{20}\log\frac{2V}{2V-32}$$

On solving V = 24 ml

49. X = 4, Y = 5

50. Final product



MATHEMATICS

51. $f(2x^2 - 1) = 2x^3 f(x)$

$f(2x^2 - 1) = -2x^3 f(-x)$

$\Rightarrow 2x^3 f(x) = -2x^3 f(-x)$ $\Rightarrow$ f(x) is an odd function

hence f(0) = 0 hence its derivative is also 0.

$$(2xy^2 - y)dx + xdy = 0$$

$$\frac{dy}{dx} + 2y^2 - \frac{y}{x} = 0$$

$$-\frac{1}{y^2}\frac{dy}{dx} + \frac{1}{y}\left(\frac{1}{x}\right) = 2$$

Let $\frac{1}{y} = z$ $\quad -\frac{1}{y^2}\frac{dy}{dx} = \frac{dz}{dx}$

$$\frac{dz}{dx} + \frac{z}{x} = 2$$

If $= e^{\int \frac{1}{x}dx} = x$

$$zx = \int 2(x)\,dx = x^2 + c$$

$$\frac{x}{y} = x^2 + c$$

Point of intersection of 2x – 3y = 1 and 3x + 2y = 8 (2, 1)

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$\frac{2}{1} = 4 + c$

$c = -2$

$\frac{x}{y} = x^2 - 2$

$k + 1 = 1$ as $k = 0$

$y(1) = -1$

52. $\tan^{-1}(x+2) + \cot^{-1}\sqrt{4x+20} = \frac{\pi}{2}$

$\Rightarrow \quad x+2 = \sqrt{4x+20} \quad x = 4 \quad \Rightarrow a = 4$

$$\lim_{x\to 0} \frac{\sqrt[4]{1-x} - \frac{(1-x)^{1/x}}{e} - bx}{5x + kx^2 + x^3} \Rightarrow \lim_{x\to 0} \frac{\left(\frac{3}{4} - b\right)x - \frac{53}{96}x^2 + \ldots.}{5x + kx^2 + x^3}$$

$$\Rightarrow \frac{3}{4} = b \text{ and } k \in R$$

53. $f(p) + f(q) = 0 \Rightarrow f(p) \cdot f(q) \le 0$

when $f(p) \cdot f(q) < 0$ then exactly one root and when $f(p) = f(q) = 0$ both the roots are p and q.

54. $$\lim_{n\to\infty} \left\{ \frac{(n+1)\left(n+\frac{1}{2}\right)\left(n+\frac{1}{4}\right).....\left(n+\frac{1}{2^{n-1}}\right)}{n^n} \right\}^n$$

$$\lim_{n\to\infty} \left(1+\frac{1}{n}\right)^n \left(1+\frac{1}{2n}\right)^n \left(1+\frac{1}{4n}\right)^n ..... \left(1+\frac{1}{2^{n-1}n}\right)^n$$

$$\Rightarrow \lim_{n\to\infty} \left(1+\frac{1}{n}\right)^n \left(1+\frac{1}{2n}\right)^{\frac{2n}{2}} \left(1+\frac{1}{4n}\right)^{\frac{4n}{4}} ..... \left(1+\frac{1}{2^{n-1}n}\right)^{\frac{2^{n-1}\cdot n}{2^{n-1}}}$$

$$\Rightarrow \lim_{n\to\infty} \left(e^1 . e^{1/2} . e^{1/4} ..... e^{1/2^{n-1}}\right)$$

$$\Rightarrow e^{1+\frac{1}{2}+\frac{1}{4}+.....\infty} = e^2 = 7.387524$$

$$I = \int_1^3 -3dx + \int_1^3 \left[(x-1)^2\right]dx, \text{ Let } x-1 = t \text{ for 2nd integral}$$

$$= -6 + \int_0^1 0\,dt + \int_0^{\sqrt{2}} 1\,dt + \int_{\sqrt{2}}^{\sqrt{3}} 2\,dt + \int_{\sqrt{3}}^{2} 3\,dt = -1 - \sqrt{2} - \sqrt{3}$$

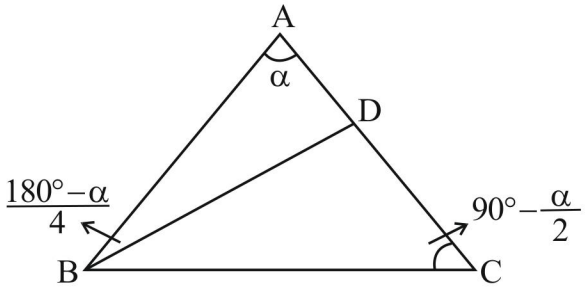
55. Let $\beta = \frac{\pi - \alpha}{4}$

Let AB = 1

Using sine rule in ΔABC and ABD (twice)

$$BC = \frac{\sin\alpha}{\sin 2\beta} \quad BD = \frac{\sin\alpha}{\sin 3\beta} \quad AD = \frac{\sin\beta}{\sin 3\beta}$$

Using BC = BD + AD



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$\sin(\pi-4\beta)\sin 3\beta = \sin(\pi-4\beta)\sin 2\beta + \sin\beta \sin 2\beta$

Using sum to product formula

$\cos\beta - \cos 7\beta = \cos 2\beta - \cos 6\beta + \cos\beta - \cos 3\beta$

$\cos 3\beta - \cos 7\beta = \cos 2\beta - \cos 6\beta$

$\sin 2\beta \sin 5\beta = \sin 2\beta \sin 4\beta$

$\Rightarrow \sin 5\beta = \sin 4\beta$

$9\beta = \pi ; \ \beta = \frac{\pi}{9}$

Hence $\beta = \frac{180^\circ - \alpha}{4}; \ \alpha = 100^\circ$

56. From f(1), f(2), f(3) …… f(2025) all have two choices for f(2026) will have only one option to make entire sum odd, hence 2^{2025}

57. $a \geq 1, b \geq 2, c \geq 3, d \geq 4$

$b - a < c - b$, $2b < a + c$ and $c - b < d - c$, $2c < b + d$

a	b	c	d
1	2	4	7
1	2	4	8
1	2	4	9
1	2	5	9
2	3	5	9
2	3	5	8
3	4	6	9

58. $\frac{1-\alpha}{1+\alpha} + \frac{1-\beta}{1+\beta} + \frac{1-\gamma}{1+\gamma} = 2\left(\frac{1}{\alpha+1} + \frac{1}{\beta+1} + \frac{1}{\gamma+1}\right) - 3$

$P(x) = x^3 - x - 1$, $P(x-1) = (x-1)^3 - (x-1) - 1$

$= x^3 - 3x^2 + 2x - 1$ has roots $\alpha + 1, \beta + 1, \gamma + 1$

Sum of reciprocals of the roots $x^3 + c_2x^2 + c_1x + c_0$ is

$-\frac{c_1}{c_0}$ so given expression is $2(2) - 3 = 1$

59. $\vec{v}_1 \cdot \vec{v}_2 = 2(\sin\alpha + \cos\alpha)\sin\beta + \cos\beta$

$|\vec{v}_1|^2 \cdot |\vec{v}_2|^2 \cos^2\theta = 9 \quad \theta = (\vec{v}_1 \wedge \vec{v}_2)$

$\Rightarrow (5 + 4\sin 2\alpha)\cos^2\theta = 9 \quad \Rightarrow \cos^2\theta = \frac{9}{5 + 4\sin 2\alpha}$

$5 + 4\sin 2\alpha \geq 9$

$4\sin 2\alpha \geq 4$

$\Rightarrow \sin 2\alpha = 1$

$\alpha = \frac{\pi}{4}$

$\cos^2\theta = 1$

$\theta = 0^\circ$

$\Rightarrow \vec{v}_1$ and $\vec{v}_2$ are collinear

$\frac{2(\sin\alpha + \cos\alpha)}{\sin\beta} = \frac{1}{\cos\beta} \qquad \alpha = \frac{\pi}{4}$

$\tan\beta = 2\sqrt{2}$

$3\tan^2\alpha + 4\tan^2\beta = 3 + 32 = 35 \qquad k_1 = 35$

$6^2 = \frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2 = \frac{9 + k^2}{10} - \left(\frac{9+k}{10}\right)^2 < 10$

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$$k < \frac{10\sqrt{10}}{3} + 1$$

$k_2 = 11$

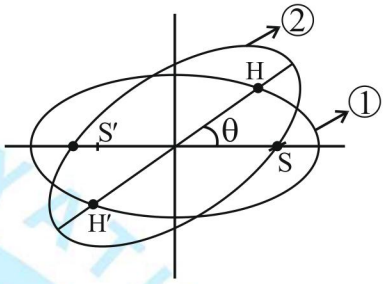
60. SS' are foci for ellipse $H'H$ are foci of ellipse 2nd HH' and SS' have same mid point

Let ellipse (1) is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ H lies on it

$(ae_2 \cos\theta, ae_2 \sin\theta)$

$$\Rightarrow \cos^2\theta = \frac{1}{e_1^2} + \frac{1}{e_2^2} - \frac{1}{e_1^2 e_2^2}$$

61.



$$S = \sum_{k=1}^{k=2024} \frac{k+2}{k! + (k+1)! + (k+2)!} = \sum_{k=1}^{2024} \frac{k+2}{k!(k+2)^2}$$

$$= \sum_{k=1}^{2024} \frac{1}{k!\,(k+2)}$$

$$= \sum_{k=1}^{2024} \frac{k+1}{(k+2)!} = \sum_{k=1}^{2024} \left[\frac{1}{(k+1)!} - \frac{1}{(k+2)!} \right] = \frac{1}{2} - \frac{1}{2026!}$$

62. Let x be the fraction

$\Rightarrow y = x - x^n$ has to be maximized

$$\frac{dy}{dx} = 0 \qquad x = \left(\frac{1}{n}\right)^{\frac{1}{n-1}} \quad \text{and} \quad \frac{d^2y}{dx^2} < 0$$

63. Let $Z = a + ib$

$$\text{Re}\left((75+117i)Z + \frac{96+144i}{Z} \right) = 81a - 108\,b = 27\,(3a - 4b)$$

to maximize $3a - 4b$

Let $x = 3a - 4b$

$$a = \frac{x+4b}{3}$$

Given $|Z| = 4$

$a^2 + b^2 = 16$

$$\Rightarrow \frac{x^2 + 8bx + 16b^2}{9} + b^2 = 16$$

$\Rightarrow 25b^2 + 8xb + (x^2 - 144) = 0$

b must be real

$$\Rightarrow (8x)^2 - 4(25)(x^2 - 144) \geq 0$$

$$-36x^2 + 14400 \geq 0$$

$$x^2 \leq 400 \qquad x \leq 20$$

So, $27(3a - 4b) = 27 \times 20 = 540$

$5 + 4 + 0 = 9$ Ans.

64. Let t be the no. of subset containing element 1

$$x = \frac{k(k-1)m}{n(n-1)} \Rightarrow t = \frac{(n-1)x}{k-1} = \frac{(n-1)k(k-1)m}{(k-1)n(n-1)} = \frac{mk}{n}$$

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65. $\frac{9}{8}+y=A\left(x-\frac{1}{4}\right)^2$

$y=A\left(x^2-\frac{x}{2}+\frac{1}{16}\right)-\frac{9}{8}$

$y=Ax^2-\frac{Ax}{2}+\frac{A}{16}-\frac{9}{8}$

$\Rightarrow$ on comparing with $y = ax^2 + bx + c$

$A=a,\ -\frac{A}{2}=b,\ \frac{A}{16}-\frac{9}{8}=C$

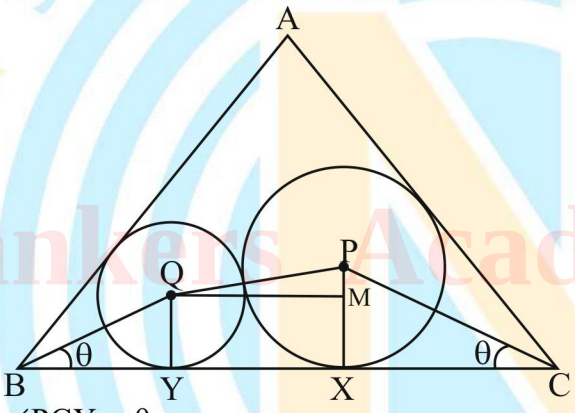
$\frac{9A-18}{16}=a+b+c$

$9A - 18$ must be divisible by 16

$9A = Z$ $Z - 18$ is divisible by 16 if $Z = 2$

$9A = 2$ $A=\frac{2}{9}$ hence $p + q = 11$

66.



$\angle QBX = \angle PCY = \theta$

$\tan 2\theta=\frac{2\tan\theta}{1-\tan^2\theta} \Rightarrow \tan\theta=\frac{3}{4}$

$XC=\frac{64}{3} \Rightarrow BY=\frac{4r}{3}$

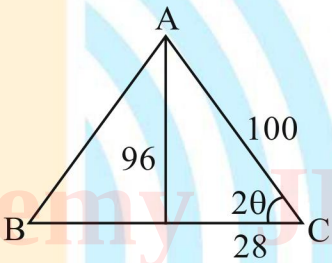
$XY=56-\left(\frac{4r+64}{3}\right)=\frac{104-4r}{3}$

QPM is a right-angled triangle

$(r+16)^2=(r-16)^2+\left(\frac{104-4r}{3}\right)^2$

$r=44-6\sqrt{35}$

$ABC \Rightarrow AB = AC$



$\tan 2\theta=\frac{24}{7}$

67. $f(x)+5\le f(x+5)\le f(x+4)+1$

$f(x+4)+1\le f(x+3)+2$

$f(x+3)+2\le f(x+2+1)+2\le f(x+2)+3\le f(x+1+1)+3\le f(x+1)+4\le f(x)+5$

$\Rightarrow f(x+1)=f(x)+1$ $f(1)=1$ $f(2)=2\ldots..$ $f(2026) = 2026$

$g(x)=f(x)+1-x$ $g(2026)=f(2026)+1-2026 = 1$

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68. $T_n = 2^{n-1}$

$$S_n = \frac{1}{2}\times 2^{n-1}\times\left(\frac{1}{10}+\frac{1}{10^2}+.....\frac{1}{10^{n-1}}\right)+2^{n-1}\times\frac{1}{10^n}$$

$$= 2^{n-1}\times\frac{1}{18}\times\left(1-\frac{1}{10^{n-1}}\right)+2^{n-1}\times\frac{1}{10^n}$$

$$\lim_{n\to\infty}\frac{S_n}{T_n} = \frac{1}{18}\left(1-\frac{1}{10^{n-1}}\right)+\frac{1}{10^n} = \frac{1}{18} \qquad a+b=19$$

69. At first obstacle $P(A_1) = 1-\frac{2}{6} = \frac{2}{3}$

At second obstacle by getting 2, 3 and 4 Jaspreet can not cross

$x + y = 2$, $x + y = 3$, $x + y = 4$ will have no. ${}^1C_1$, ${}^2C_1$ and ${}^3C_1$ +ve integral solution

${}^1C_1 + {}^2C_1 + {}^3C_1 = 6$ so $P(A_2) = 1-\frac{6}{6^2} = \frac{5}{6}$

$P(A_3)$ $x+y+z=k$ $\quad k\in\{3, 4, 5, 6, 7, 8\}$ $\quad 1\le x, y, z\le 6$

$=1+3+6+10+15+21=56$

$$P(A_3) = 1-\frac{56}{6^3} = \frac{20}{27}$$

$$P(A_1).P(A_2).P(A_3) = \frac{2}{3}\times\frac{5}{6}\times\frac{20}{27} = \frac{100}{243}$$

70. $|\overrightarrow{AB}|^2 + |\overrightarrow{CD}|^2 = 3^2+11^2 = 130$

$|\overrightarrow{BC}|^2 + |\overrightarrow{DA}|^2 = 7^2+9^2 = 130$

Since $\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CD}+\overrightarrow{DA}=0$

$$|\overrightarrow{DA}|\cdot|\overrightarrow{DA}| = (\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CD})\cdot(\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CD})$$

$$|\overrightarrow{DA}|^2 = |\overrightarrow{AB}|^2+|\overrightarrow{BC}|^2+|\overrightarrow{CD}|^2+2(\overrightarrow{AB}\cdot\overrightarrow{BC}+\overrightarrow{BC}\cdot\overrightarrow{CD}+\overrightarrow{AB}\cdot\overrightarrow{CD})$$

$$|\overrightarrow{DA}|^2 = |\overrightarrow{AB}|^2-|\overrightarrow{BC}|^2+|\overrightarrow{CD}|^2+2\left(|\overrightarrow{BC}|^2+\overrightarrow{AB}\cdot\overrightarrow{BC}+\overrightarrow{BC}\cdot\overrightarrow{CD}+\overrightarrow{CD}\cdot\overrightarrow{AB}\right)$$

$$|\overrightarrow{DA}|^2 = |\overrightarrow{AB}|^2-|\overrightarrow{BC}|^2+|\overrightarrow{CD}|^2+2(\overrightarrow{AB}+\overrightarrow{BC})\cdot(\overrightarrow{BC}+\overrightarrow{CD})$$

$$\Rightarrow 2|\overrightarrow{AC}|\cdot|\overrightarrow{BD}| = 2|\overrightarrow{AB}+\overrightarrow{BC}|\cdot|\overrightarrow{BC}+\overrightarrow{CD}|$$

$= AD^2+BC^2-AB^2-CD^2 = 0$

Thus $|\overrightarrow{AC}|\cdot|\overrightarrow{BD}|$ has only one value

71. Let $b_n = \frac{1}{a_n}$

$$\left(3-\frac{1}{b_{n+1}}\right)\left(6+\frac{1}{b_n}\right) = 18$$

$3b_{n+1}-6b_n-1=0$

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$b_{n+1} = 2b_n + \frac{1}{3}$ or $\left(b_{n+1} + \frac{1}{3}\right) = 2\left(b_n + \frac{1}{3}\right)$

So, $\left\{b_n + \frac{1}{3}\right\}$ is a GP with common ratio 2

$b_n + \frac{1}{3} = 2^n\left(b_0 + \frac{1}{3}\right) = 2^n\left(\frac{1}{a_0} + \frac{1}{3}\right)$ $(a_0 = 3)$

$b_n = \frac{2^{n+1}}{3} - \frac{1}{3}$

$\sum_{i=0}^{2026} \frac{1}{a_i} = \sum_{i=0}^{2026} b_i = \frac{1}{3}\sum_{i=0}^{2026}\left(2^{i+1} - 1\right)$

$= \frac{1}{3}\left[\frac{2\left(2^{n+1}-1\right)}{2-1} - (n+1)\right] = \frac{1}{3}\left(2^{n+2} - n - 3\right)$

$\sum_{i=0}^{2026} = \frac{1}{3}\left(2^{2028} - (2028+1)\right)$

$k_1 = 3$ and $k_2 = 2028$ $3 \times 2028 = 6084$

72. $u = \frac{4}{4-x^2} + \frac{9}{9-y^2} = \frac{4}{4-x^2} + \frac{9x^2}{9x^2-1}$

$= 1 + \frac{35x^2}{-9x^4 + 37x^2 - 4} = 1 + \frac{35}{37 - \left(\left(3x - \frac{2}{x}\right)^2 + 12\right)}$

Minima occurs at $x = \sqrt{\frac{2}{3}}$

Value is $\frac{12}{5}$

73. $\int_0^4 f(x)dx - \int_0^4 g(x)dx = 10$

$\left(A_1 + A_3 + A_4\right) - \left(A_2 + A_3 + A_4\right) = 10$

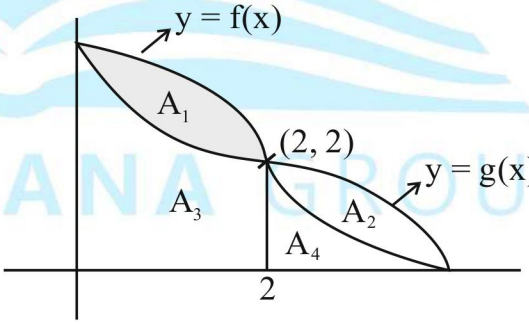
$A_1 - A_2 = 10$

$\int_2^4 g(x)dx - \int_2^4 f(x)dx = 5$

$\left(A_2 + A_4\right) - A_4 = 5$

$A_2 = 5$

$A_1 = 15$



74. $\vec{r} = bj + \lambda(bj - ck)$ $\vec{r} = ai + u(ai + ck)$

Vector perpendicular to both

$= \begin{vmatrix} i & j & k \\ 0 & b & -c \\ a & 0 & c \end{vmatrix} = bci - acj - abk$

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Shortest distance

$$2d = \left|\frac{(ai - bj)\cdot(bci - acj - abk)}{\sqrt{b^2c^2 + a^2c^2 + a^2b^2}}\right|$$

$$2d = \frac{2abc}{\sqrt{a^2b^2 + b^2c^2 + c^2a^2}}$$

$k = 1$

$$\left|Adj(Adj\,A)\right| = \begin{vmatrix} 14 & 18 & -14 \\ -14 & 14 & 28 \\ 28 & -14 & 14 \end{vmatrix} = 14\times14\times14 \begin{vmatrix} 1 & 2 & 1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{vmatrix}$$

$(14)^3\,[3 - 2\,(-5) - 1\,(-1)] = 14^4$

$|A|^4 = (14)^4 \quad |A| = 14$

75. $$I = \int_0^1 \frac{(2x+1)x + 2 + 2x + 2x^2}{2\sqrt{1 + x + x^2}}dx$$

Let $I_1 = \int_0^1 \frac{(2x+1)x}{2\sqrt{1+x+x^2}}dx$

Integrating by parts

$$I_1 = x\sqrt{1+x+x^2}\Big]_0^1 - \int_0^1 \sqrt{1+x+x^2}\,dx$$

$$I = I_1 + \int_0^1 \sqrt{1+x+x^2}\,dx$$

$$I = x\sqrt{1+x+x^2}\Big|_0^1 = \sqrt{3}$$

$I^2 = 3$

$I^2 + 19 = 22$

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